

Coherence assist, reduced

Dirichlet–Gershgorin is too crude, the two-period is a global AP, and the signed CD off-diagonal is the remainder

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Abstract

Round 385 reduced geometric quietness $Q_k < 1$ to the pair (Δ, C) , and named C as the r285 factor $\lambda_{\max}(E_k) = \text{maxdiag}_k \cdot (1 + \text{assist}_k)$. This note asks whether assist_k admits a source-defined upper bound, from ν -angle geometry on the cosine grid, that pushes $\lambda_{\max} < 1$ through $k = N - 1$. What is proved: the Chebyshev– T Christoffel–Darboux kernel is the Dirichlet combination $[D_{k-1}(\theta - \phi) + D_{k-1}(\theta + \phi)]/(2\pi)$; the bookkeeping identity; the Gershgorin majorant; the cosine-mesh identity $\Delta_{\text{mesh}} = \pi/S$; and that the two-period adversary is a *single arithmetic progression* of ν -angles ($\rho_{\text{AP}} = 1$). What is *not* proved: Gershgorin plus the 3/8-floor closes the budget $\text{assist} < (1 - \text{maxdiag})/\text{maxdiag}$. On FRAME-A $w = 9$, Gershgorin crosses 1 at $k = 21$ while $n_0 = \lfloor 2N/5 \rfloor = 73$ and the wall sits at $k = 183$; core-42 has $k_{\text{Gershgorin}}^*/N \leq 0.114 < 2/5$. The 3/8-floor does not upgrade ν -angle gaps: $d_{\min} = 1$ mesh step and $d_{\text{med}} = 3$ mesh steps on $w = 9$, so $d_{\min}/d_{\text{med}} = 1/3 < 3/8$. The hydra-legal remainder is the *signed* μ -CD off-diagonal (true assist 0.040 at the $w = 9$ wall against Gershgorin assist 13.3, cancellation 99.7%). No RH claim.

Status. Lemmas 1–4 are proved (finite identities). Lemma 5 is a construction identity of the two-period toy. Proposition 1 is a finite machine census. Proposition 2 names the reduction. No RH claim. No L^* claim.

1 The object

Write B_k for the ν -dressed matrix of the first k μ -orthonormal polynomials and $E_k = B_k B_k^T$, so

$$(E_k)_{ij} = \sqrt{v_i v_j} K_k(y_i, y_j), \quad (1)$$

with K_k the μ -Christoffel–Darboux kernel of degree k . The r285 identity

$$\lambda_{\max}(E_k) = \text{maxdiag}_k \cdot (1 + \text{assist}_k), \quad \text{assist}_k = \lambda_{\max}/\text{maxdiag}_k - 1 \quad (2)$$

is exact. L^* is $\lambda_{\max}(E_{N_w}) < 1$. Round 382 reduced a flank-mass entry to $n_0 = \lfloor 2N/5 \rfloor$ and named this assist as the L^2 remainder for $n_0 = N - 1$. Round 385 showed $Q_k < 1$ is not sufficient (two-period $c = 2/3$).

A parenthetical correction: r385 quoted “assist 1.5–4” at $k = 183$ against $\max Q \approx 0.39$. That mixed the *single-degree* sampling $Q_k = \langle \pi_k^2 \rangle_\nu$ with $\text{maxdiag}(E_k) = \max_j v_j K_k(y_j, y_j)$. The wall values are $\text{maxdiag}_{183} = 0.9614$ and $\text{assist}_{183} = 0.0399$, inside the r285 ladder 0.0059–0.0702.

The candidate of this round: a geometric bound on the off-diagonal of (1) from Dirichlet evaluations at ν -angle differences, using the 3/8-floor and F1 run-lengths, feeding Gershgorin or Schur so that $\lambda_{\max} < 1$ through $k = N - 1$.

2 What the grid sees

Write $D_n(\alpha) = \sin((n + \frac{1}{2})\alpha) / \sin(\alpha/2)$ for the Dirichlet kernel ($D_n(0) = 2n + 1$).

Lemma 1 (Chebyshev–Dirichlet). *For the Chebyshev- T orthonormal system on $[-1, 1]$ ($p_0 = \pi^{-1/2}$, $p_n = \sqrt{2/\pi} T_n$),*

$$K_n(\cos \theta, \cos \phi) = \frac{D_{n-1}(\theta - \phi) + D_{n-1}(\theta + \phi)}{2\pi}.$$

This is an identity of trigonometric polynomials, independent of any window. It is the Chebyshev proxy of (1), not the μ -CD kernel.

Lemma 2 (bookkeeping). *Identity (2) holds for every Hermitian E with positive diagonal: it is the definition of assist. On the 2×2 equality case $E = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ one has $\lambda_{\max} = 3$, $\text{maxdiag} = 2$, $\text{assist} = \frac{1}{2}$.*

Lemma 3 (Gershgorin). $\lambda_{\max}(E) \leq \max_i (E_{ii} + \sum_{j \neq i} |E_{ij}|)$. *On the 2×2 of Lemma 2 this is equality (the all-ones mode). The same number is the Schur / ℓ^∞ bound. It uses absolute off-diagonals.*

Lemma 4 (cosine mesh). *On the construction grid $x_j = \cos(\pi j/S)$ the angles are $\theta_j = \pi j/S$ and consecutive mesh steps equal π/S . A ν -run of length 2 (permitted by F1) realises $d_{\min}(\nu) = \pi/S$. The 3/8-floor of the log-sieve controls fold gaps, not this mesh step: on FRAME-A $w = 9$ one has $d_{\min} = 1$ mesh, $d_{\text{med}} = 3$ mesh, hence $d_{\min}/d_{\text{med}} = 1/3 < 3/8$.*

Lemma 5 (two-period AP). *The two-period toy of amplitude c on S nodes occupies every other cosine index. Its ν -angles form a single arithmetic progression of common difference $2\pi/S$, so $\rho_{\text{AP}} := L_{\text{AP}}/n_\nu = 1$ and consecutive gaps are unique. This is the occupation of one residue class.*

3 The geometric candidate, measured

Write k_G^* for the last degree with Gershgorin majorant < 1 , and $\rho_{\text{AP}} = L_{\text{AP}}/n_\nu$ for the longest arithmetic progression in the ν -angle set, divided by the number of ν -atoms.

Proposition 1 (census). *On the sealed constructors:*

1. *FRAME-A $w = 9$: $N = 184$, $n_\nu = 104$, $k_G^* = 20$ (first crossing $k = 21$), $n_0 = 73$, $k_G^*/N = 0.109$. At $k = 183$: $\lambda_{\max} = 0.999830$, $\text{maxdiag} = 0.9614$, $\text{assist} = 0.0399$, Gershgorin assist 13.32, cancellation $1 - \text{assist}/A_G = 0.997$. $\rho_{\text{AP}} = 0.0769$, $\text{gap-CV} = 0.813$, $d_{\min}/d_{\text{med}} = 1/3$. Dirichlet max-row at $k = 22$: 0.291.*
2. *Core-42: $\rho_{\text{AP}} \in [0.034, 0.168] < 1/5$; $\text{gap-CV} \in [0.536, 0.892] > 1/2$; $k_{\text{Gershgorin}} \in [8, 32]$, all strictly below n_0 , and $k_G/N \leq 0.114 < 2/5$. The mutant bar $\rho_{\text{AP}} \leq 0.03$ fails (max 0.168).*
3. *EXT-heavy seven $\{69, 96, 97, 99, 107, 117, 129\}$: $\rho_{\text{AP}} \in [0.006, 0.033] < 0.05$, $\text{gap-CV} \geq 0.56$, and $\lambda_{\max}(E_{80}) < 1$ (sharpest 0.9965 at $k=97$).*

Gershgorin does not improve the r382 entry $n_0 = \lfloor 2N/5 \rfloor$. The unsigned Dirichlet row-sum grows with k and crosses 1 near $0.11 N$.

The 3/8-floor cannot feed a better $d_{\min}$: F1 already permits a run of two, which saturates the mesh. A crude Dirichlet cap $1/\sin(d_{\min}/2) \sim 2S/\pi$ is $\Theta(N)$ and useless past $k \sim 10$.

4 Why the two-period dies

The two-period of amplitude $c = 2/3$ on $S = 81$ has $Q \equiv \alpha = 0.683 < 1$ and $\lambda_{\max}(E_{22}) = 1.029 > 1$. Its ν -atoms are equidistant: $\rho_{\text{AP}} = 1$, gap-CV 0.078 (wrap residual), Dirichlet max-row at $k = 22$ equal to 0.858, about three times the source. Dirichlet peaks add. Amplitude $c = 1$ dies at $k = 1$ ($\lambda_{\max} = 1.025$).

The source is not a residue class. Log-near prime powers plus F1 (run-lengths ≤ 2) produce a spread of consecutive ν -gaps ($w = 9$: multipliers 0.375, 0.625, 1, 1.375, ... in eighths of the median; twelve distinct values against the two-period's one). That is the named property:

$$\rho_{\text{AP}} < \frac{1}{5} \quad \text{on canonical windows.} \quad (3)$$

It is construction-level (the fold image of the prime-power comb is not one residue class) and is census-grade on core-42 and the EXT-heavy seven. It is *not* sufficient for $\lambda_{\max} < 1$: scramble seed 1 has $\rho_{\text{AP}} = 0.064$ like the source and still crosses $\lambda_{\max} \geq 1$ at $k = 22$ (maxdiag crosses 1 later, at $k = 50$). Scramble is a diagonal / product kill, not an AP kill.

F1 alone does not kill the two-period: the two-period has max-run 1. F1 does kill the clustered run-of-3 ($k_\lambda = 15 < n_0 = 16$ on $S = 81$, $\rho_{\text{AP}} = 1$). A clustered run-of-2 has $k_\lambda = 21 > n_0$ — F1 permits it, and the r382 entry survives.

5 The reduction

Proposition 2 (reduction). *The candidate “assist $_k \leq (1 - \text{maxdiag})/\text{maxdiag}$ from 3/8-separation + Gershgorin” is strictly larger than the signed remainder:*

1. Chebyshev–Dirichlet CD, bookkeeping, Gershgorin, mesh. SATZ (Lemmas 1–4).
2. Two-period = global AP. SATZ (Lemma 5), with Dirichlet max-row $\sim 3 \times$ the source at the r385 witness degree $k = 22$.
3. Unsigned geometry does not close. $k_G^* \sim 0.11 N < \lfloor 2N/5 \rfloor$ on every tested window; $d_{\min}/d_{\text{med}} = 1/3 < 3/8$.
4. Named property $\rho_{\text{AP}} < 1/5$: kills the two-period, holds on EXT-heavy seven, holds on scramble (which dies elsewhere).
5. Remainder, strictly smaller: the signed μ -CD off-diagonal — the true assist $_k$, whose unsigned majorant is closed as too crude. At the $w = 9$ wall this is a 0.040 spectral excess against a 13.3 Gershgorin excess. An assist theorem that would dock to *ChristoffelPivotBound* (r384) has to bound this signed object. $n_0 = \lfloor 2N/5 \rfloor$ is not improved; $n_0 = N - 1$ stays open with that remainder.

No Lean file is edited in this round. The r384 named Prop *ChristoffelPivotBound* already accepts a Rayleigh bound on E_{k+1} ; a proved assist estimate would be a hypothesis of that Prop, not a new sorry.

6 Kill battery

world	ρ_{AP}	first $\lambda_{\text{max}} \geq 1$	notes
MAIN $w = 9$	0.077	none through $N - 1$	$k_G = 21$
core 42	≤ 0.168	none through k_G	$k_G \leq 32$
EXT-heavy 7	≤ 0.033	none at depth 80	property holds
scramble $w = 9$	0.064 (holds)	$k = 22$	maxdiag at 50
two-period $c = 2/3$	1 (breaks)	$k = 21$	$Q = 0.683 < 1$
two-period $c = 1$	1	$k = 1$	mass
clustered run 3	1	$k = 15 < n_0$	F1 forbids
clustered run 2	1	$k = 21 > n_0$	F1 allows
mutant $\rho \leq 0.03$	—	—	fails on core

7 Machine checks

Numbered lemmas: `verify_coherence_assist.py` (same directory). Standalone: Chebyshev–Dirichlet, 2×2 bookkeeping and Gershgorin, two-period AP, mesh, $1/3 < 3/8$. Construction: FRAME-A $w = 9$ pins, Dirichlet max-row, two-period λ_{max} , scramble, $c = 1$, clustered run-3. Discovery census: `coherence_assist_probe.py` (`--smoke 14/14`, full 20/20). Exit line: COHERENCE ASSIST VERIFIED.

Claim boundary

Finite identities for the Chebyshev–Dirichlet kernel, the Gershgorin majorant, and the cosine mesh; a named reduction of the geometric assist candidate to the signed μ -CD off-diagonal; a scramble break on the diagonal; a two-period witness that global AP occupation, not $Q < 1$, is the coherence kill. No statement about zeros of $\zeta(s)$, in either direction. No L^* claim.

References

- [1] TFPT RH program, round 285: Christoffel decomposition $\lambda_{\text{max}} = \text{maxdiag} \cdot (1 + \text{assist})$.
- [2] S. Hamann, *The entry lemma of the pivot band*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.
- [3] RH/FlankEntry.lean, round 384: ChristoffelPivotBound named.
- [4] S. Hamann, *Christoffel quietness, reduced*, standalone note, August 2026, `rh/problem/christoffel_quiet.tex`.
- [5] S. Hamann, the 3/8-floor of the mean sieve, campaign probe `mean_sieve_floor_probe.py`.